Inorganic Cume
March 6, 2009

Photoredox reactions often form the basis of solar energy conversion schemes. In general terms, or using a specific example, answer the following questions:

(1) What kind(s) of excited states (d-d, MLCT, LMCT, etc.) typically undergo photoredox reactions?

(2) Explain how the electronic redistribution in these excited states results in useful redox properties.

(3) How do the following parameters enter into the design of solar energy conversion catalysts?

a) luminescence lifetime

b) intersystem crossings efficiency

c) absorption spectrum

d) redox potentials

e) radiationless decay rate